School of Computing

Human-Machine Interaction and Deciding Whether or Not to Treat.

Using Large Language Models to Support Decision-Making in Kidney Transplant.

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Introduction

This interdisciplinary research project aims to address a rapidly growing field of computer science research (Large Language Models) and evaluate the potential of such new and evolving technologies in kidney transplant, exploring practical and ethical requirements from a human-machine interaction perspective.

Background

The decision-making process for kidney transplant surgery is complex, involving multiple stages influenced by human and technical factors, but ultimately, the majority of smaller decisions are made by a human clinician, unsupported by technology.

Since the release of OpenAI's ChatGPT in November 2022, Large Language Models (LLMs) have gained popularity due to their user-friendly interface and potential in medical applications, such as clinical notes management, patient care chatbots, and diagnostics [1].

This research aims to understand the decision process as a decision ecology, identify factors contributing to uncertainty, and explore how LLMs can manage these factors while ensuring explainability and managing ethical concerns both generally, and those specific to healthcare.

Sample

Involvement and inclusion/exclusion criteria for human participants involved in all research phases are shown in Table 1.

Participant Samples	Number of Participants	Interviews per Participant	Inclusion Criteria	Exclusion Criteria
Non-NHS Professionals	Approx. 8	1	• Non-NHS staff with specialised technical knowledge on either decision-support systems, AI and automation in healthcare, or AI decision systems.	• Individuals without relevant technical knowledge/expertise. • Individuals under the age of 18 • Anyone unable to provide informed consent to participate in the study.
Initial Interviews	Approx. 12	1 (60 minutes each)	• NHS staff working in a role with knowledge or expertise of kidney transplant, involvement with patient care, or contact with kidney transplant patients, or knowledge of advanced transplant technology or decision support systems. • Age 18 or over and able to consent. • Fluent in English language.	• Non-NHS staff • NHS staff without specific knowledge and experience in kidney transplant. • Individuals without relevant knowledge/expertise. • Individuals under the age of 18 • Anyone unable to provide informed consent to participate in the study.
Feedback Interviews	Approx. 6	3 (30 minutes each)		

Table 1. Study inclusion/exclusion criteria.

Aims

- Explore the relationship between human autonomy and machine learning
- Understand the requirements for an AI-based decision system in kidney transplant. Evaluate existing policies and requirements of ethical AI systems and translate this into the context of kidney transplant, considering the specific considerations which would need to be accounted for.
- Explore the potential for reinforcement of biases and errors when using large language models.

Methods

A combination of qualitative and quantitative methods are to be used, enabling an in-depth understanding of the current decision system and the factors influencing it, as well as testing concepts to propose a framework for an LLM-based decision-support system, in 3 overlapping phases of research.

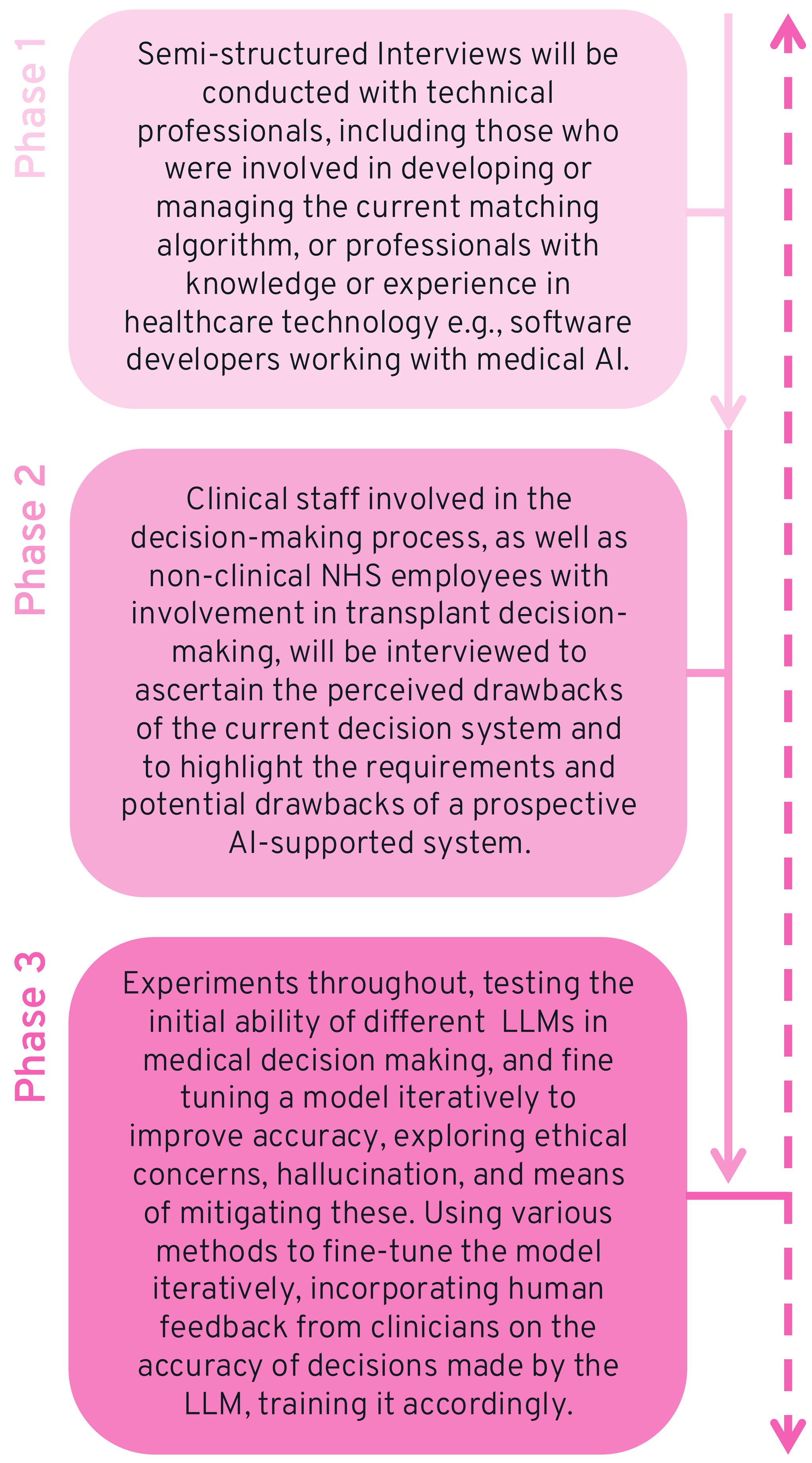

Figure 1. Project phases.

Context

Decision-making in kidney transplant is done by a human, and a matching algorithm. Figure 2 shows the current decision system, and where this project intends to explore incorporating AI decision support.

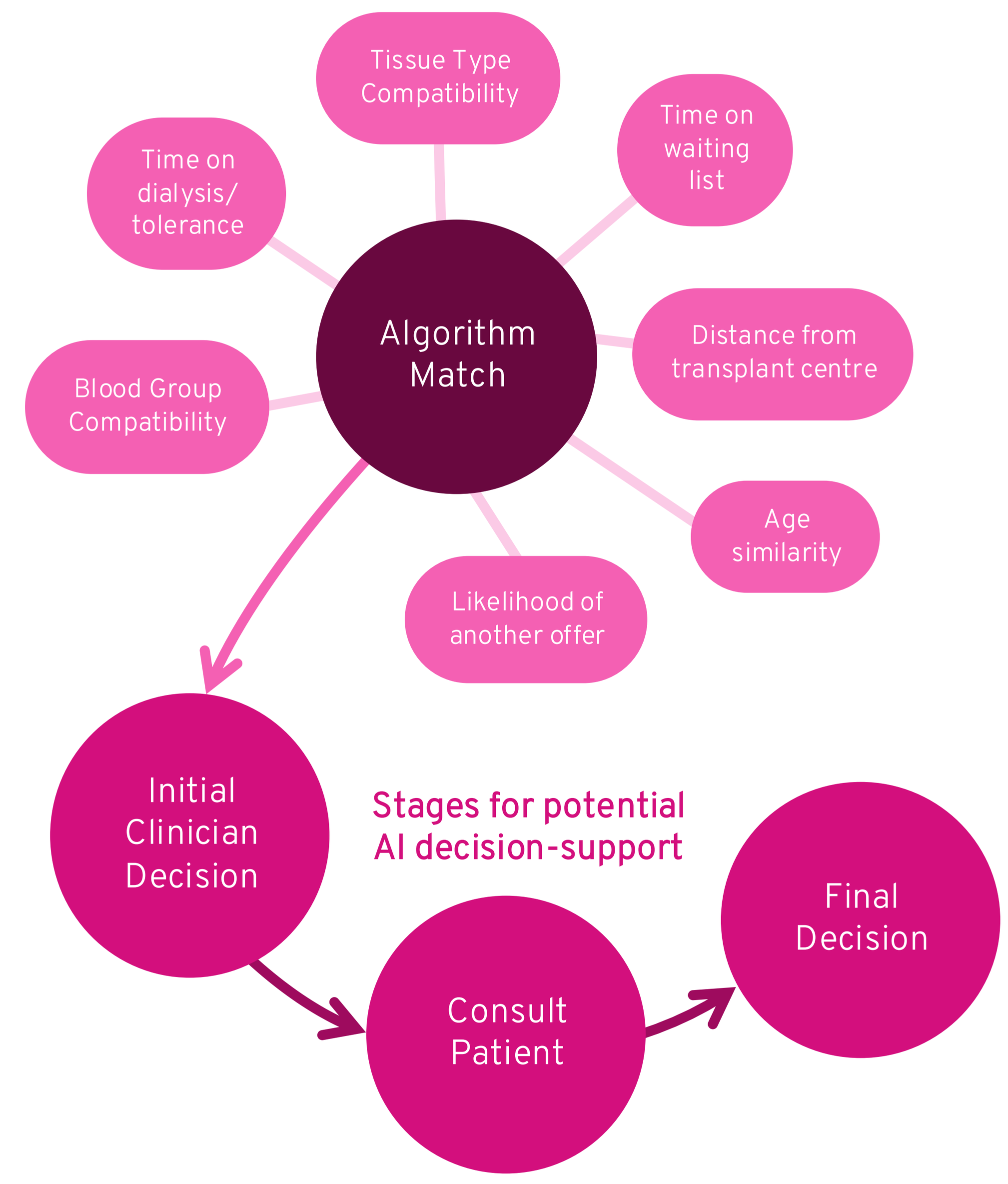

Figure 2. Current kidney transplant decision system [2].

Whilst LLMs can be trained for greater accuracy using medical information [3], their ability to respond to queries and complete tasks without being trained specifically to do so is considered to be an indicator of significant potential within medicine [4].

LLMs have shown recent success when applied to medical scenarios, specifically in clinical notes management, patient care chatbots, and even diagnosis [5] with new uses continually being acknowledged and understood through further research.

Future work

Phase 1 of research is currently in progress, with the initial setup for Phase 3 also ongoing. Phase 2 will begin on the approval of NHS ethics applications, and most activities in Phase 3 will follow.

References

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